

Coin Detection and Classification Pipeline

Chosen Field and Motivation

This project falls under the field of **computer vision and machine learning**, with a focus on **automated object detection and classification**. The motivation is to develop a low-cost, scalable system capable of identifying and classifying coins based on their physical features for applications such as vending machines, coin sorters, or educational tools.

Dataset and Why It Was Selected

A custom dataset of coin images was used, including various denominations and colors (silver and copper). This dataset was chosen to:

- Match real-world lighting and background conditions.
 - Provide diverse orientations, lighting, and overlap scenarios.
 - Focus specifically on color and size—key attributes for classification.
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1. Preprocessing and Image Enhancement

Input: Raw coin image (RGB format)

Output: Enhanced grayscale image with improved contrast and reduced noise

Tasks Implemented:

- **Image Resizing:** Normalizes image dimensions (1024 px width).
- **Color Space Conversion:** BGR to grayscale.
- **Contrast Adjustment:** CLAHE enhances regional contrast.
- **Noise Reduction:** Gaussian blur smooths the image.

Justification:

CLAHE works better for non-uniform lighting compared to global histogram equalization. Gaussian Blur reduces edge noise for better segmentation.

2. Segmentation

Input: Preprocessed grayscale image

Output: Isolated regions (ROIs) representing individual coins

Tasks Implemented:

- **Hough Circle Transform:** Detects circular shapes corresponding to coins.

Justification:

Hough Transform is robust for detecting circular features and does not require prior knowledge of coin positioning.

3. Feature Extraction

Input: Cropped coin regions (ROI)

Output: Normalized feature vector for each coin

Tasks Implemented:

- **Color Histogram:** 3D RGB histogram (8x8x8 bins)
- **Normalization:** Feature vectors are scaled for consistent input.

Justification:

Color histogram effectively distinguishes material (e.g., copper vs. silver), which is a key classification cue.

4. Classification

Input: Normalized feature vectors

Output: Predicted material class (Copper or Silver)

Tasks Implemented:

- **MLPClassifier (Multi-Layer Perceptron):** A neural network-based classifier.

Justification:

MLP is capable of modeling complex, non-linear relationships between color features and coin material.

🏆 5. Evaluation and Performance Metrics

Input: Predicted vs. actual labels

Output: Numerical performance metrics

Metrics Used:

- **Train/Test Accuracy:** Basic performance indicator.
- **Precision, Recall, F1-Score:** (Suggested for further analysis)

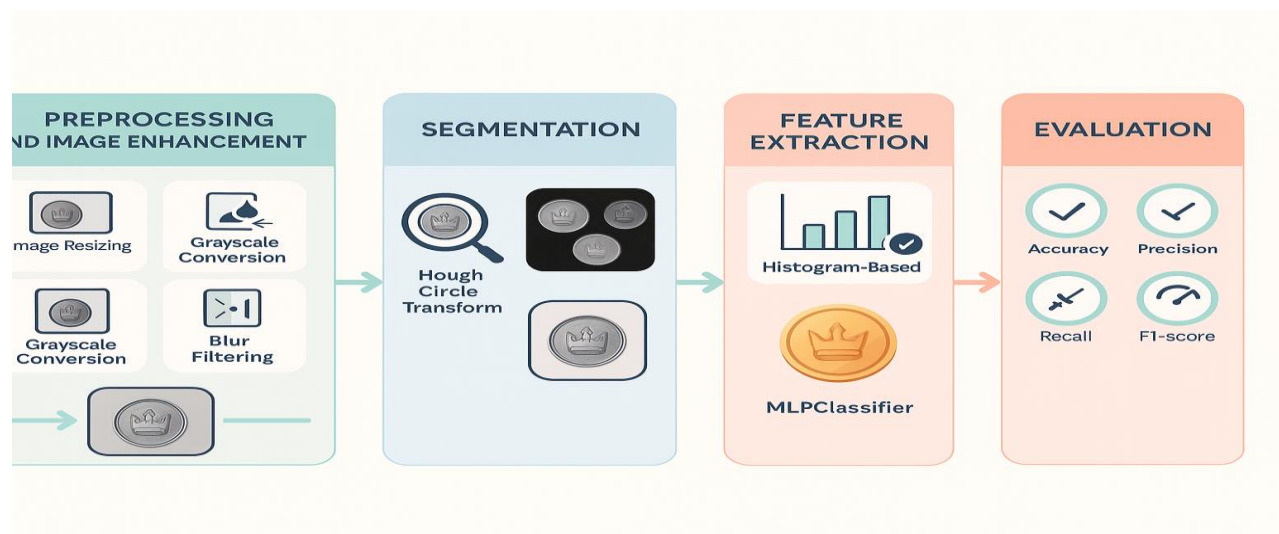
Discussion:

High training accuracy was observed. For deeper analysis, confusion matrices and additional metrics (like F1-score) should be computed.

🖼️ Visual Examples of Detection Output

Below is a visual representation of the pipeline:

- Preprocessed input image
- Hough Circle Detection overlay
- ROI-based classification labels



Limitations or Challenges Faced

- Sensitive to poor lighting and shadows
 - Overlapping coins may affect segmentation
 - Feature-based methods limited in generalization compared to CNNs
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References

- OpenCV Documentation: <https://docs.opencv.org/>
- Scikit-learn Documentation: <https://scikit-learn.org/>
- Digital Image Processing Textbooks and Tutorials